
MB 555: Deep Learning for Business Applications

Course Code	MB 555
Course Name	Deep Learning for Business Applications
L-T-P-C	2-0-0-2
Intended for	MBA
Mutual Exclusion	NA
Approval	54th BoA (May 2024)

Course Contents

- **Deep Learning Basics:** History of Deep learning, Mathematics behind deep learning, activation functions, and their derivatives, Loss functions, Deep networks, and fully connected networks, variations of gradient descent, Regularization in Networks. (4 Hours)
- **Autoencoders:** Encoding, layer-wise pretraining, types of auto-encoders, Autoencoders for finance applications. (4 Hours)
- **Convolutional Neural Networks:** Convolution basics, filters, padding, multidimensional convolution, and convolution layers, pooling, striding, and transposed convolution, Applying filters for image-identification-related applications. (6 Hours)
- **Recurrent Neural Networks:** Recurrent neural network basics, dealing with language, training recurrent neural networks, long short-term memory, RNN for forecasting. (4 Hours)
- **Reinforcement Learning:** Basics of reinforcement learning and its structure, the multi-arm bandit problem, the game of Nim, Temporal difference learning, and Q-learning. (6 hours)
- **GAN:** Introducing GAN, forging concepts, forging with neural networks. (4 Hours)
- **Presentation:** (2 hours)

Text Books

1. NA

References

1. J. Krohn, G. Beyleveld and A. Bassens, **Deep Learning Illustrated**, Pearson India Education Services Ltd, 2020.
2. Nikhil Buduma and Nicholas Locascio, **Fundamentals of Deep Learning: Designing Next- Generation Machine Intelligence Algorithms**, Shroff Publishers, 2017.
3. Ian Goodfellow, Yoshua Benagio and Aaron Courville, **Deep Learning**, MIT Press, October 2016.
4. Armando Vieira and Bernardete Ribeiro, **Introduction to Deep Learning Business Applications for Developers: From Conversational Bots in Customer Service to Medical Image Processing**, Apress, 2018.
5. Eugene Charniak, **Introduction to Deep Learning**, MIT Press, 2019
6. Terrence J. Sejnowski, **The Deep Learning Revolution**, MIT Press, 2019
7. M Gopal, **Deep Learning**, Pearson India Education Services Ltd, 2022